



Science — Cbse Annual Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	Local winds like the mountain breeze influence weather conditions and daily life in mountainous regions. Which of the following is a benefit of these local winds? (A) They increase atmospheric CO₂ levels. (B) They accelerate the melting of glaciers. (C) They cause ocean currents to reverse direction. (D) They regulate temperature and moisture conditions, supporting soil and crop health. Answer: Correct Option: (D) ► Why? D. Local winds in mountainous regions help regulate temperature and moisture conditions, and they support soil and crop health. [Ch 13: Earth as a System: Energy, Matter, and Life]	1
2	When two forces are opposite in direction but unequal in magnitude, what is the magnitude of the net force? (A) The sum of the magnitudes of the two forces (B) The difference between the magnitudes of the two forces (C) The average of the magnitudes of the two forces (D) Zero Answer: Correct Option: (B) ► Why? B. When two forces are opposite in direction but unequal in magnitude, the magnitude of the net force is equal to the difference between the magnitudes of the two forces. [Ch 6: How Forces Affect Motion]	1



Q.NO.	QUESTIONS	MARKS
3	Who proposed that any living plant cell can develop into a complete plant if provided with suitable nutrients and favourable conditions? (A) Arun Kumar Sharma (B) Theodor Schwann (C) Rudolf Virchow (D) Gottlieb Haberlandt Answer: Correct Option: (D) ► Why? D. In 1902, Austrian botanist Gottlieb Haberlandt proposed the idea of totipotency in plant cells. [Ch 2: Cell: The Building Block of Life]	1
4	In a game of carrom, a moving striker hits a stationary white coin. What happens to the energy of the white coin? (A) It decreases because the striker does negative work on it (B) It increases because the striker does positive work on it (C) It remains constant because no work is done (D) It becomes zero upon collision Answer: Correct Option: (B) ► Why? B. The striker applies force in the direction of displacement of the white coin, doing positive work and thereby increasing its energy. [Ch 7: Work, Energy, and Simple Machines]	1
5	What are the special organelles found in plant cells that include chloroplasts, leucoplasts, and chromoplasts? (A) Plastids (B) Lysosomes (C) Ribosomes (D) Endoplasmic reticulum Answer: Correct Option: (A) ► Why? A. Plant cells contain special organelles called plastids, such as chloroplasts, leucoplasts, and chromoplasts. [Ch 2: Cell: The Building Block of Life]	1
6	Assertion (A): The formation of different types of tissues leads to division of labour in multicellular organisms. Reason (R): Different groups of cells perform different functions, which increases the efficiency of the body and enables it to carry out complex life processes. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (A) ► Why? A. The assertion is true because the formation of different tissues leads to division of labour, which increases efficiency. The reason correctly explains this by stating that different groups of cells performing different functions enable the body to carry out complex life processes. [Ch 3: Tissues in Action]	
7	Assertion (A): The divisions of science into branches like physics, chemistry, biology and earth science are independent of each other. Reason (R): Most real-world problems today, such as understanding climate change, developing medicines, or designing sustainable technologies, require ideas from several disciplines together. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? D. The assertion is false because the source explicitly states that these divisions are not independent of each other—they are made by us only to help organise knowledge. The reason is true, as real-world problems like understanding climate change, developing medicines, and designing sustainable technologies do require ideas from several disciplines together, which actually demonstrates why the branches are interconnected rather than independent. [Ch 1: Exploration: Entering the World of Secondary Science]	1
8	Which joint allows forward, backwards, sideways, and circular movements? (A) Hinge joint (B) Pivot joint (C) Fixed joint (D) Ball and socket joint Answer: Correct Option: (D) ► Why? D. The ball and socket joint allows forward, backwards, sideways and circular movements. [Ch 3: Tissues in Action]	1
9	The claim that food becomes harmful during an eclipse is best disproved by asking which type of question? (A) Whether people have believed this claim for centuries (B) Whether the eclipse is total or partial (C) Whether any physical, chemical, or biological mechanism supports the claim (D) Whether ancient texts mention this practice Answer: Correct Option: (C) ► Why? C. Disproof comes from asking simple scientific questions. Since an eclipse is just a play of shadows, no physical, chemical, or biological mechanism supports such a claim.	1



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	
10	After a zygote is formed, it undergoes a series of mitotic divisions while travelling to the uterus and then: (A) Is released through the vagina (B) Returns to the ovary (C) Degenerates within a day (D) Implants into the inner lining of the uterus Answer: Correct Option: (D) ► Why? D. The zygote undergoes mitotic divisions while travelling to the uterus and implants into the inner lining of the uterus to receive nourishment, marking the beginning of pregnancy. <i>[Ch 11: Reproduction: How Life Continues]</i>	1
11	Which of the following is an example of a colloid? (A) Milk (B) Salt water (C) Sand in water (D) Sugar in water Answer: Correct Option: (A) ► Why? A. Milk is identified as a colloid in the source material. Salt water and sugar in water are solutions, while sand in water is a suspension. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
12	What is the primary role of the dermal tissue system in plants? (A) Transporting water and minerals throughout the plant (B) Forming the main body of the plant between dermal and conducting tissues (C) Forming the outer covering of the plant, protecting inner parts and reducing water loss (D) Increasing the length and girth of the plant Answer: Correct Option: (C) ► Why? C. The dermal tissue system forms the outer covering of the plant, protects the inner parts, and reduces water loss. <i>[Ch 3: Tissues in Action]</i>	1
13	Given that the atomic mass of K = 39 u and Cl = (A) 35.5 u (B) 39 u (C) 3.5 u (D) 74.5 u Answer: Correct Option: (D)	1



Q.NO.	QUESTIONS	MARKS
	► Why? D. The formula unit mass of $KCl = 39\text{ u} + 35.5\text{ u} = 74.5\text{ u}$. [Ch 9: Atomic Foundations of Matter]	
14	Brass is an alloy composed of approximately: (A) 80% zinc and 20% copper (B) 50% copper and 50% zinc (C) 80% copper and 20% zinc (D) 80% copper and 20% tin Answer: Correct Option: (C) ► Why? C. Brass is described as a mixture of approximately 80% copper and 20% zinc in the source material. [Ch 5: Exploring Mixtures and their Separation]	1
15	Which of the following statements about loudness and intensity is correct? (A) Loudness and intensity are exactly the same quantity (B) Intensity is a measurable quantity, whereas loudness depends on the listener's hearing ability (C) Intensity is subjective, while loudness is measurable (D) Loudness is independent of amplitude Answer: Correct Option: (B) ► Why? B. Intensity is a measurable physical quantity, whereas loudness depends on the listener's hearing ability. Although they are often used interchangeably in everyday language, they are not the same. [Ch 10: Sound Waves: Characteristics and Applications]	1
16	In the airplane fuel miscalculation incident, what was the cause of the error? (A) The fuel tank had a leakage mid-flight (B) The pilot misread the fuel gauge instruments (C) The density of fuel was used in pounds per litre instead of kilograms per litre (D) The aircraft was overloaded with passengers and cargo Answer: Correct Option: (C) ► Why? C. The ground crew miscalculated the fuel required since they used the density of fuel in pounds (lb) per litre rather than kilograms (kg) per litre. [Ch 1: Exploration: Entering the World of Secondary Science]	1



Q.NO.	QUESTIONS	MARKS
17	Assertion (A): The sound of submerged metal spoons tapped under water reaches the listener after travelling through both water and air. Reason (R): Sound can only travel through a single medium at a time. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. The assertion is true because the source explicitly states that the sound of the submerged spoons has reached the listener after travelling through water and air. The reason is false because sound can travel through multiple media sequentially, as demonstrated by the fact that the sound passes through both water and air before reaching the listener, contradicting the claim that sound can only travel through a single medium at a time. [Ch 10: Sound Waves: Characteristics and Applications]	1
18	How does the elongated, cylindrical body structure of roundworms benefit them? (A) It allows them to fly efficiently (B) It helps them produce their own food (C) It reduces water loss in dry environments (D) It allows efficient movement through soil, water, or host tissues Answer: Correct Option: (D) ► Why? D. The elongated, cylindrical body structure allows roundworms efficient movement through soil, water, or host tissues. [Ch 12: Patterns in Life: Diversity and Classification]	1
19	Assertion (A): The ovary of a flower develops into a fruit after fertilisation. Reason (R): The ovules inside the ovary develop into seeds. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? Both the assertion and reason are true statements based on the events following fertilisation, but the reason does not explain why the ovary develops into a fruit. The ovary developing into a fruit and the ovules developing into seeds are concurrent outcomes of fertilisation, not a cause-and-effect relationship between the two. [Ch 11: Reproduction: How Life Continues]	1



Q.NO.	QUESTIONS	MARKS
20	Assertion (A): Most cellular activities in prokaryotic cells take place directly in the cytoplasm. Reason (R): Prokaryotic cells lack membrane-bound organelles. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. The assertion is true because prokaryotic cells perform their cellular activities directly in the cytoplasm. The reason correctly explains this by stating that prokaryotic cells lack membrane-bound organelles, which means there are no separate compartments for these activities, forcing them to occur in the cytoplasm. [Ch 2: Cell: The Building Block of Life]	1
Section B Attempt 5 out of 7.		
21	What is the purpose of using models in science, and what do they involve? Answer: Models are simplified ways of looking at real systems that focus only on what is most important for a given question. Building models involves making assumptions and deliberately ignoring certain details to keep things simple enough while still allowing us to find answers. ► Why? This tests understanding of the concept of scientific models, their purpose, and the deliberate choices involved in constructing them, as described in the source. [Ch 1: Exploration: Entering the World of Secondary Science]	2
	OR In a simple pendulum, why does the bob eventually stop oscillating instead of reaching the same height indefinitely? Answer: The pendulum bob eventually stops because of energy loss due to friction at the support and air resistance. These external forces cause the mechanical energy of the system to decrease over time, preventing it from reaching the same height. ► Why? This applies the concept of energy dissipation to the real-world observation of a pendulum slowing down. [Ch 7: Work, Energy, and Simple Machines]	
22	Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+}. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. Answer: Cl^-, one, two	2



Q.NO.	QUESTIONS	MARKS
	► Why? According to the textbook context, a chlorine atom gains one electron to acquire a negative charge, becoming a chloride anion represented as Cl^-. Since a magnesium atom gives two electrons to become Mg^{2+}, and each chlorine atom can only accept one electron, it requires two chlorine atoms to accept the two electrons. Therefore, one ion of magnesium and two ions of chlorine combine to form magnesium chloride, ensuring that the positive and negative charges balance each other to create a neutral overall structure. ► Step-by-step: - Step 1: Identify the ion formed by chlorine. The context states chlorine gains one electron to become a chloride anion, Cl^-. - Step 2: Determine the ratio of ions. Magnesium forms a Mg^{2+} ion by giving two electrons. Since each Cl atom takes only one electron, two Cl^- ions are needed to accept the two electrons from one Mg^{2+} ion. - Step 3: Fill in the blanks with 'Cl^-', 'one', and 'two'. [Ch 9: Atomic Foundations of Matter]	
	OR How does the magnitude of the force of friction affect the distance travelled by an object before coming to rest? Answer: When the force of friction is smaller, the velocity of the object decreases more slowly and it travels a larger distance before coming to rest. A larger force of friction causes the velocity to decrease faster, so the object travels a shorter distance before stopping. ► Why? The source concludes from Activities 6.1 and 6.2 that smaller friction leads to slower velocity decrease and larger distance travelled. [Ch 6: How Forces Affect Motion]	
23	Why are the epithelial tissues lining an animal's internal organs usually only one or a few cells thick? Answer: Epithelial tissues are typically only one or a few cells thick to allow the quick exchange of materials across them. This thinness facilitates efficient diffusion and transport, which is essential for their protective and exchange functions. ► Why? The text explicitly states that the thinness of epithelial tissues allows for the quick exchange of materials across them, ruling out other potential reasons like strength or friction reduction. [Ch 3: Tissues in Action]	2
24	Define isotonic, hypotonic, and hypertonic solutions based on solute concentration. Answer: An isotonic solution has equal solute concentration outside and inside the cell. A hypotonic solution has lower solute concentration outside than inside. A hypertonic solution has higher solute concentration outside than inside the cell. ► Why? The source defines each type by comparing the solute concentration of the extracellular medium to the intracellular medium. [Ch 2: Cell: The Building Block of Life]	2



Q.NO.	QUESTIONS	MARKS
25	Differentiate between sublimation and deposition based on the state changes involved. Answer: Sublimation is the process where a solid changes directly into the vapour state without passing through the liquid state upon heating. Deposition is the reverse process where vapours condense back into a solid without becoming a liquid on cooling. ► Why? The source defines sublimation as a solid-to-vapour change without becoming liquid, and deposition as vapour condensing back to a solid without becoming liquid. [Ch 5: Exploring Mixtures and their Separation]	2
Section C Attempt 5 out of 7.		
26	A perfume mixture is prepared by mixing 15 mL of fragrance oil with enough water to make a total volume of 75 mL of solution. Calculate the volume by volume percentage (% v/v) of the fragrance oil in the mixture. Answer: The volume by volume percentage (% v/v) indicates the millilitres of solute present in 100 millilitres of the solution. It is calculated using the formula: % v/v = (Volume of solute / Volume of solution) × 100. Here, the volume of fragrance oil (solute) is 15 mL, and the total volume of the solution is 75 mL. Substituting the values in the formula: % v/v = (15 mL / 75 mL) × 100. Calculation: (1 / 5) × 100 = 20%. Therefore, the perfume mixture contains 20% v/v of fragrance oil in the solution. ► Why? The calculation uses the formula for volume by volume percentage explicitly provided in the text, applying it to the given quantities of solute and solution. ► Step-by-step: - Step 1: Identify the formula for volume by volume percentage: % v/v = (Volume of solute / Volume of solution) × 100. - Step 2: Note the given values: Volume of fragrance oil = 15 mL, Volume of solution = 75 mL. - Step 3: Substitute the values into the formula and calculate: (15/75) × 100 = 20% v/v. [Ch 5: Exploring Mixtures and their Separation]	3
	OR Explain how the structure of the inner membrane of a mitochondrion is well-adapted to facilitate energy production.	



Q.NO.	QUESTIONS	MARKS
	Answer: Mitochondria are known as the powerhouses of the cell because they supply energy for cellular activities. Each mitochondrion is surrounded by two membranes: an outer smooth membrane and an inner membrane. The inner membrane is uniquely folded into finger-like projections called cristae. These folds are crucial because they increase the surface area for the chemical reactions involved in energy production. During cellular respiration, glucose and other molecules are broken down in the mitochondria to release energy. This energy is stored as Adenosine Triphosphate (ATP), which acts as the energy currency for the cell. ► Why? The cristae's folded structure maximizes surface area on the inner mitochondrial membrane, directly optimizing the space available for the chemical reactions of cellular respiration and ATP production. ► Step-by-step: 1. Step 1: Identify the inner membrane's folded structures as cristae. 2. Step 2: Explain that cristae increase the surface area for chemical reactions. 3. Step 3: Connect this to cellular respiration, where glucose breakdown releases energy stored as ATP. [Ch 2: Cell: The Building Block of Life]	
27	State the principle of the lever using the relationship between effort, load, and their respective arms. How does increasing the effort arm affect the effort required? Answer: A lever is a rigid bar that rotates about a fixed point called the fulcrum. It works on the principle that the work done at one end is transferred to the other end. The relationship is given by $\text{Effort} \times \text{Effort arm} = \text{Load} \times \text{Load arm}$. The effort arm is the distance of the effort from the fulcrum, and the load arm is the distance of the load from the fulcrum. By increasing the effort arm, the lever applies a larger force to the load than the applied effort. Therefore, increasing the effort arm reduces the effort required to move the same load, making the task easier. ► Why? A lever balances the moments of effort and load; increasing the effort arm reduces the effort needed to lift the same load. ► Step-by-step: 1. Step 1: State the lever principle: $\text{Effort} \times \text{Effort arm} = \text{Load} \times \text{Load arm}$. 2. Step 2: Define effort arm and load arm as distances from the fulcrum. 3. Step 3: Conclude that increasing the effort arm reduces the effort required to move the same load. [Ch 7: Work, Energy, and Simple Machines]	3
	OR Describe the arrangement of ions in a sodium chloride crystal and explain what a crystal lattice is.	



Q.NO.	QUESTIONS	MARKS
	Answer: Ionic compounds usually do not remain as single units but form three-dimensional (3-D) crystals with ions arranged in a repeating pattern. In sodium chloride (NaCl), each Na⁺ ion is surrounded by six Cl⁻ ions, and each Cl⁻ ion is surrounded by six Na⁺ ions. This regular, repeating 3-D arrangement is called a crystal structure. The crystal structure is represented as a crystal lattice, with ions depicted as points or dots. This representation helps visualise the arrangement of ions in the crystal. ► Why? In NaCl crystals, ions are arranged in a regular 3-D repeating pattern where each ion is surrounded by six oppositely charged ions; this arrangement is depicted as a crystal lattice. ► Step-by-step: 1. Step 1: Ionic compounds form 3-D crystals with ions in a repeating pattern. 2. Step 2: In NaCl, each Na⁺ is surrounded by 6 Cl⁻ ions and vice versa — this is the crystal structure. 3. Step 3: The crystal lattice represents this structure with ions shown as points or dots. [Ch 9: Atomic Foundations of Matter]	
28	An aircraft ran out of fuel mid-flight because the ground crew used the density of fuel in pounds per litre instead of kilograms per litre. What does this incident highlight about the importance of using standard units (SI)? Answer: This incident highlights that using standard units (SI) everywhere avoids conversions and errors that can lead to serious consequences. Pounds and kilograms are very different units, and mixing them up caused the aircraft to be 15,000 litres short of fuel. Standard units are based on agreed international standards rather than local objects or opinions. They allow scientific results to be compared universally and ensure fairness in daily life and trade. Without standard units, measurements would be confusing and unreliable, making safe operations and accurate communication impossible. ► Why? The incident demonstrates that failing to use standard international units leads to conversion errors, whereas SI units ensure consistency, safety, and fairness. ► Step-by-step: 1. Step 1: Identify the error caused by mixing pounds and kilograms, leading to a fuel shortage. 2. Step 2: Explain how using standard units everywhere avoids such dangerous conversion errors. 3. Step 3: Mention that standard units allow comparison of scientific results and ensure fairness in trade. [Ch 1: Exploration: Entering the World of Secondary Science]	3
29	Differentiate between the apical, lateral, and intercalary meristems based on their location and function in plants.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Plants have three types of meristematic tissues that contribute to growth in different ways. The apical meristem is located at the root and shoot tips, and its primary function is to increase the length of the plant. The lateral meristem is located along the circumference of stems and is responsible for increasing the girth of the plant. The intercalary meristem is located at the base of certain plants, such as grasses, and helps them regenerate after cutting. Together, these meristems account for growth in length, girth, and branching in plants. ► Why? The three meristems are distinguished by their specific locations (tips, circumference, base) and their respective growth functions (length, girth, regeneration). ► Step-by-step: 1. Step 1: Identify the apical meristem at root and shoot tips, which increases length. 2. Step 2: Identify the lateral meristem along the circumference of stems, which increases girth. 3. Step 3: Identify the intercalary meristem at the base of grasses, which helps in regeneration. [Ch 3: Tissues in Action]	
30	According to Newton's second law of motion, how does the acceleration of an object relate to the net force acting on it and its mass? Answer: Newton's second law of motion states that when a net force acts on an object, the object accelerates in the direction of the net force. The magnitude of the acceleration is directly proportional to the magnitude of the net force applied. This means if the force increases, the acceleration increases proportionally. Additionally, the acceleration is inversely proportional to the mass of the object. This means that for the same net force, an object with a larger mass will experience a smaller acceleration than an object with a smaller mass. ► Why? The answer outlines the two key proportionalities of Newton's second law: acceleration is proportional to net force and inversely proportional to mass, as explicitly stated in the source. ► Step-by-step: 1. Step 1: Identify the direction of acceleration relative to the net force. 2. Step 2: State the direct proportionality of acceleration magnitude to the net force. 3. Step 3: State the inverse proportionality of acceleration to the object's mass. [Ch 6: How Forces Affect Motion]	3
Section D Attempt 4 out of 6.		
31	Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess.	5



Q.NO.	QUESTIONS	MARKS
	Answer: In science, laws, theories, and principles are terms with specific and distinct meanings that help organise our understanding of the world in a systematic way. A law usually describes a regular pattern observed in nature, often expressed using words or mathematical relationships. For example, Newton's laws of motion explain the jerk felt when a bus stops suddenly. Laws describe what happens based on repeated observations. A theory goes a step further than a law. It provides an explanation of why those patterns occur, usually based on evidence gathered over time and available at that time. For example, the atomic theory explains how molecules are formed. In science, a theory does not mean a guess or an untested idea; it is an explanation based on careful testing and critical examination. Principles are broad ideas that help us make sense of a given situation. For example, the principle of conservation of energy can be applied when climbing up the stairs. Principles serve as guiding ideas rather than descriptions or explanations. A key feature that makes science reliable is that these ideas are always open to improvement and often change as new evidence becomes available. No scientific theory is ever final and none is beyond question. This openness to being corrected by nature itself is what allows science to help us understand the world we live in. ► Why? Laws describe observed patterns, theories explain why patterns occur based on evidence, and principles are broad guiding ideas. A scientific theory is a well-tested explanation, not a guess. ► Step-by-step: 1. Step 1: Define a law and give the Newton's laws of motion example. 2. Step 2: Define a theory and give the atomic theory example; clarify it is not a guess. 3. Step 3: Define a principle and give the conservation of energy example. 4. Step 4: Explain how all these ideas remain open to improvement as new evidence emerges. 5. Step 5: Conclude by stating why this openness makes science reliable. [Ch 1: Exploration: Entering the World of Secondary Science]	
	OR Describe the process of meiosis and explain its significance in living organisms. Where does meiosis occur in animals and plants, and how is the original chromosome number restored after meiosis?	



Q.NO.	QUESTIONS	MARKS
	Answer: Meiosis is a type of cell division that produces gametes and occurs only in the cells of reproductive organs. Gametes produced through meiosis create variations and diversity among living organisms, which is why children resemble their parents but are not exactly the same. In animals, including humans, meiosis occurs in the testes of males to produce sperm and the ovaries of females to produce eggs. In plants, meiosis occurs in the anthers (male parts) to form pollen grains (that later produce sperm cells) and the ovaries (female parts) to produce egg cells. During meiosis, the parent cell divides twice, one after the other, to form four daughter cells. In the first division, the cell divides into two daughter cells and the number of chromosomes in each daughter cell is reduced to half. The second division is similar to mitosis, where each daughter cell divides into two, thus forming four daughter cells, each with half the number of chromosomes. As a result, each gamete has half the number of DNA compared to the parent cell. During fertilisation, when gametes from two individuals combine, the original chromosome number is restored. The significance of meiosis lies in producing gametes for sexual reproduction, which generates variations and diversity among living organisms, ensuring that offspring are not identical to their parents. ► Why? Meiosis is a two-step reductional division producing four haploid gametes from one diploid parent cell, ensuring genetic variation and restoring the diploid state upon fertilisation. ► Step-by-step: 1. Step 1: Define meiosis as a cell division producing gametes in reproductive organs. 2. Step 2: State where meiosis occurs in animals (testes, ovaries) and plants (anthers, ovaries). 3. Step 3: Describe the two-step division process — first division halves chromosomes, second is like mitosis. 4. Step 4: Explain that each gamete has half the DNA and that fertilisation restores the original chromosome number. 5. Step 5: Conclude with the significance — variations, diversity, and resemblance to parents. [Ch 2: Cell: The Building Block of Life]	
32	How would you separate the mixtures given in Table 5.3? Mention the reason for choosing your method. If a mixture cannot be separated, explain why.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Mud from muddy water can be separated by filtration or sedimentation because suspended particles settle in muddy water over time. Plasma from other components in blood cannot be separated by these simple methods because, unlike muddy water, the particles in such mixtures do not settle over time. Naphthalene and sand are separated by sublimation. Naphthalene undergoes a transition directly into a vapour without passing through the liquid state, leaving sand behind. Chalk powder and common salt are separated by filtration and crystallization. When water is added, salt dissolves, chalk is filtered out, and salt is obtained by crystallization. Common salt and water are separated by distillation or crystallization. Distillation obtains a liquid from a mixture of a liquid and a solid, while crystallization gets a pure solid from its saturated solution. Oil from water is separated using a separating funnel because they are immiscible liquids that form separate layers based on their different densities. Pigments of a flower are separated by paper chromatography, which separates compounds by taking advantage of differences in their rates of movement on the paper. ► Why? The answer maps each mixture to a separation technique explicitly defined in the context. Mud settles, so sedimentation/filtration applies. Blood plasma separation isn't covered by the text's simple methods since its particles don't settle like mud. Naphthalene sublimates. Chalk and salt use dissolution, filtration, and crystallization. Salt and water use distillation or crystallization. Oil and water are immiscible, requiring a separating funnel. Flower pigments use paper chromatography based on movement rates. [Ch 5: Exploring Mixtures and their Separation]	
	OR A cricket fielder gradually pulls their hands backwards while catching a fast-moving ball, and airbags are provided in vehicles. Explain how Newton's second law of motion is applied in both these situations to reduce the force experienced. Also, explain why a coconut brought down at high velocity against a hard surface experiences a very large force.	



Q.NO.	QUESTIONS	MARKS
	Answer: Newton's second law of motion establishes that force is proportional to acceleration. This means that for a given change in velocity, if the time duration over which this change occurs is increased, the magnitude of acceleration decreases, and consequently the force required also decreases. While catching a fast-moving ball in cricket, the fielder gradually pulls their hands backwards with the moving ball just after catching it. By doing so, the fielder increases the time duration during which the high velocity of the ball reduces to zero. This reduces the magnitude of the acceleration of the ball as it slows down, thus requiring a smaller force to be applied by the fielder to stop the ball. Applying a smaller force also minimises injury to the fielder. Airbags in vehicles work on a similar principle. In the event of a collision, the vehicle comes to an abrupt halt, and the airbag inflates quickly into a soft compressible cushion. Instead of directly hitting the hard steering wheel or dashboard, the passenger's head and chest push into the bag. This increases the time over which the hitting occurs, resulting in a smaller acceleration and hence a reduced force exerted on the person, thereby lowering the risk of serious injuries, particularly when combined with seat belt usage. In contrast, when a fresh coconut is brought down at a very high velocity to hit a hard surface, it stops in a very short time. To change its velocity so quickly, the ground must exert a very large force on it. This large force breaks the shell of the coconut, demonstrating that a shorter time interval for velocity change leads to a much larger force. ► Why? Increasing the time over which velocity changes reduces acceleration and hence force (Newton's second law); decreasing the time (as with the coconut) increases the force dramatically. ► Step-by-step: 1. Step 1: State the core principle from Newton's second law linking force, acceleration, and time 2. Step 2: Explain the cricket fielder's action — pulling hands back increases time, reduces acceleration, reduces force 3. Step 3: Explain airbags — soft cushion increases contact time, reduces acceleration, reduces force on passenger 4. Step 4: Explain the coconut — very short stopping time causes very large acceleration and very large force 5. Step 5: Conclude by contrasting the two strategies (increasing vs. decreasing time) and their effects on force [Ch 6: How Forces Affect Motion]	
33	Describe the structural characteristics and types of meristematic tissues in plants. Explain how these tissues give rise to permanent tissues.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Meristematic tissue is a group of cells that undergo continuous and rapid cell division in plants. The cells are small, have thin cell walls, a large and prominent nucleus, and dense cytoplasm with many organelles. Vacuoles are generally absent, and the cells are tightly packed with little or no intercellular space. There are three types of meristematic tissues based on their location. Apical meristem is located at the root and shoot tips and increases the length of the plant. Lateral meristem is located along the circumference of stems and increases the girth. Intercalary meristem is located at the base of certain plants, such as grasses, and helps them regenerate after cutting. Due to continuous cell division, meristematic tissue adds new cells to the plant body. Some newly formed cells remain meristematic, while others lose the ability to divide. These cells undergo changes in their structure and function to become specialised for specific functions like support, transport, or storage. This process, by which meristematic tissue becomes specialised to perform specific functions, is called differentiation. Through differentiation, the cells lose their ability to divide and become permanent tissues, contributing to the division of labour in the plant body. ► Why? Meristematic tissues are actively dividing cells responsible for plant growth, characterized by specific structural features like dense cytoplasm and lacking vacuoles. They are classified by location into apical, lateral, and intercalary types. When these cells stop dividing and specialize, they form permanent tissues through differentiation. ► Step-by-step: 1. Step 1: Define meristematic tissue and describe its cellular characteristics (small size, thin walls, prominent nucleus, dense cytoplasm, absent vacuoles, tightly packed). 2. Step 2: List and explain the three types of meristematic tissues: apical, lateral, and intercalary, along with their locations and functions. 3. Step 3: Explain what happens when meristematic cells lose their ability to divide. 4. Step 4: Define the process of differentiation and how it leads to the formation of permanent tissues. [Ch 3: Tissues in Action]	
34	Describe the energy transformations that take place in a traditional water mill (gharat) used in the Himalayan region to grind grain. How is this same principle applied in modern times to generate electricity?	5



Q.NO.	QUESTIONS	MARKS
	Answer: In the Himalayan region, a traditional water mill called the gharat or panchakki is used to grind grain. This device demonstrates the conversion of potential energy into kinetic energy and then into useful work, illustrating the principle that energy can change from one form to another. The water starts from the top of a hill possessing potential energy due to its height. As the water flows down through a pipe, this potential energy gets converted to kinetic energy — the energy possessed by the water due to its motion as it descends. The kinetic energy of the flowing water drives a wheel and sets it into rotational motion. This rotating wheel is connected to a grinding stone at the top, which grinds the grain. Thus, the mechanical energy of the water is ultimately used to perform the work of grinding. In modern times, the same principle is applied on a much larger scale. The potential energy of water stored in dams is similarly converted into kinetic energy as the water flows down, and this kinetic energy is then used to generate electricity. Diagram description: Draw a hillside with a pipe (labelled A) carrying water from the top. At the bottom of the pipe, show a water wheel (labelled B) being driven by the flowing water. Connected to the wheel, show a grinding stone at the top (labelled C). Label the potential energy at the top of the pipe and kinetic energy at the bottom. In both the gharat and modern dams, the fundamental energy transformation is the same: potential energy of water at a height converts to kinetic energy of moving water, which then performs useful work. ► Why? <i>The gharat and modern dams both rely on the conversion of gravitational potential energy of water at a height into kinetic energy, which is then harnessed to do useful work — grinding grain or generating electricity.</i> ► Step-by-step: 1. Step 1: <i>Introduce the gharat as a traditional water mill in the Himalayan region.</i> 2. Step 2: <i>Explain that water at the top has potential energy, which converts to kinetic energy as it flows down the pipe.</i> 3. Step 3: <i>Describe how the kinetic energy drives the wheel into rotational motion, which turns the grinding stone.</i> 4. Step 4: <i>Draw and label a diagram showing the pipe (A), wheel (B), and grinding stone (C).</i> 5. Step 5: <i>Explain the modern application — dams convert potential energy of stored water to kinetic energy to generate electricity.</i> <i>[Ch 7: Work, Energy, and Simple Machines]</i>	
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Observing Cells Under a Microscope Rohan observed onion peel cells under a microscope. He counted 25 cells along the diameter of the visible field, which was 5000 μm. His microscope had an eyepiece and an objective lens, each with a magnifying power of 10X. He noticed the cell membrane formed a distinct boundary around each cell. He recalled that the cell membrane is composed of proteins and lipids, and that it serves as the boundary where substances move between the cell and its external environment. He also learned that microscopes are improved by enhancing three main features: resolution, contrast, and magnification. (i) Calculate the estimated size of one onion peel cell based on the data provided. (ii) What is the total magnification of Rohan's microscope? How many times larger will the cell appear under this magnification? (iii) The cell membrane is composed of proteins and lipids. Describe the role of the cell membrane as a boundary between the cell and its external environment. (iv) Among the three main features of a microscope—resolution, contrast, and magnification—which one is described as the 'measure of clarity,' and why is improving this feature important for studying cell structures? Answer: (i) Estimated size of one onion peel cell = $5000 \mu\text{m} / 25 = 200 \mu\text{m}$. (ii) Total magnification = $10X \times 10X = 100X$. The cell will appear 100 times larger. (iii) The cell membrane serves as the boundary where substances move between the cell and its external environment. It enables the cell to interact with its surroundings and exchange materials, allowing the cell to function as a unit. Even single-celled organisms exchange materials and respond to their environment through the cell membrane. (iv) Resolution is described as the measure of clarity. Improving resolution is important because it allows scientists to see very small structures clearly, making the microscope a powerful tool for studying cell structure. ► Why? i) Estimated size of one onion peel cell = $5000 \mu\text{m} / 25 = 200 \mu\text{m}$. ii) Total magnification = $10X \times 10X = 100X$. The cell will appear 100 times larger. iii) The cell membrane serves as the boundary where substances move between the cell and its external environment. It enables the cell to interact with its surroundings and exchange materials, allowing the cell to function as a unit. Even single-celled organisms exchange materials and respond to their environment through the cell membrane. iv) Resolution is described as the measure of clarity. Improving resolution is important because it allows scientists to see very small structures clearly, making the microscope a powerful tool for studying cell structure. [Ch 2: Cell: The Building Block of Life]	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: Building Scientific Models A student wants to model how long it takes for a ball thrown upward to return to the ground. She considers several factors: the mass of the ball, its initial speed and direction, air resistance, the colour of the ball, and the spin of the ball. She recalls that physicist Meghnad Saha simplified the study of stars by ignoring complex processes and focusing only on temperature, pressure, and how atoms formed ions, which helped explain the connection between star colour and temperature. She decides to apply a similar simplification approach. She also notes that science uses precise symbols like m for mass, v for velocity, and F for force, and that equations are compact statements about relationships between quantities. (i) What three factors did Meghnad Saha focus on to explain the colour of stars? (ii) Why should the student ignore the colour of the ball when modeling the time it takes to return to the ground? (iii) The student ignores air resistance and spin in her simple model. Later, she wants a more accurate prediction. What should she do, based on the approach described for building complex models? (iv) The student writes an equation relating distance, time, and velocity to describe the ball's motion. Explain why this equation is more than just a calculation tool. Answer: (i) Temperature, pressure, and how atoms formed ions. (ii) The colour of the ball does not affect the motion of the ball, just as the colour of a cricket ball makes no difference to whether it crosses the boundary. Only factors that affect the outcome should be included. (iii) She should add extra details such as air resistance and spin for greater accuracy. As we build more and more complex models, we add extra details beyond the simple model to improve the accuracy of our predictions. (iv) An equation is a compact statement about how certain things are related. Mathematics in science is a language that helps us think more clearly about the world, not just a tool for finding numerical answers. ► Why? i) Temperature, pressure, and how atoms formed ions. ii) The colour of the ball does not affect the motion of the ball, just as the colour of a cricket ball makes no difference to whether it crosses the boundary. Only factors that affect the outcome should be included. iii) She should add extra details such as air resistance and spin for greater accuracy. As we build more and more complex models, we add extra details beyond the simple model to improve the accuracy of our predictions. iv) An equation is a compact statement about how certain things are related. Mathematics in science is a language that helps us think more clearly about the world, not just a tool for finding numerical answers. [Ch 1: Exploration: Entering the World of Secondary Science]	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Concentration of Solutions in Everyday Products A student examines three products. A packaged food item contains 8 g of salt in 200 g of the product. A glucose solution bottle from a pharmacy is labelled 5% m/v, meaning 5 g of glucose is present in 100 mL of the solution. A milk powder packet lists its protein content as 15% m/m. The student notes that mass by mass percentage is commonly used for heterogeneous mixtures like milk powder, while mass by volume percentage is used in medicines and laboratories where measuring the volume of a liquid is easier than weighing it. (i) What does 5% m/v on the glucose solution bottle indicate? (ii) Why is mass by volume percentage used for medicines like glucose solution instead of mass by mass percentage? (iii) Calculate the mass by mass percentage of salt in the packaged food item if it contains 8 g of salt in 200 g of the product. (iv) Milk powder is labelled using mass by mass percentage for its composition. What type of mixture is milk powder, and why is this method of expressing concentration appropriate for it? Answer: (i) 5% m/v means that 5 grams of glucose is present in 100 millilitres of the solution. (ii) Mass by volume percentage is used because measuring the volume of a liquid is easier than weighing it in medicines and laboratories. (iii) Mass of solute (salt) = 8 g, Mass of solution = 200 g. Mass by mass percentage = (Mass of solute / Mass of solution) × 100 = (8/200) × 100 = 4% m/m. (iv) Milk powder is a heterogeneous mixture. Mass by mass percentage is appropriate because this method is also used for heterogeneous mixtures such as milk powder and spice mixtures. ► Why? i) 5% m/v means that 5 grams of glucose is present in 100 millilitres of the solution. ii) Mass by volume percentage is used because measuring the volume of a liquid is easier than weighing it in medicines and laboratories. iii) Mass of solute (salt) = 8 g, Mass of solution = 200 g. Mass by mass percentage = (Mass of solute / Mass of solution) × 100 = (8/200) × 100 = 4% m/m. iv) Milk powder is a heterogeneous mixture. Mass by mass percentage is appropriate because this method is also used for heterogeneous mixtures such as milk powder and spice mixtures. [Ch 5: Exploring Mixtures and their Separation]	5



Q.NO.	QUESTIONS	MARKS
37	Case Study: Totipotency and Tissue Culture In 1958, F.C. Steward demonstrated that phloem cells from carrot could regenerate whole plants when grown in a nutrient medium containing simple sugars and hormones. The phloem cells first dedifferentiated into an undifferentiated mass of unspecialised cells, then redifferentiated into roots, shoot, and a complete plant. His experiments showed that cells in liquid medium with nutrients and air showed 20% increase in fresh weight, while cells in solid medium under light, or in liquid medium without air (condition X), showed reduced growth. Separately, scientists found that <i>Agrobacterium tumefaciens</i> causes crown gall disease, creating tumour-like swellings from rapid, uncontrolled cell division, and this bacterium is now used to introduce useful genes into plants. (i) Define totipotency. (ii) In Steward's experiment, cells in liquid medium with air showed 20% increase in fresh weight, while cells in condition X (liquid medium with nutrients) showed reduced growth. What does condition X lack, and why did growth reduce? (iii) A scientist cultures phloem cells from a carrot plant in a nutrient medium with simple sugars, hormones, and air. Describe the complete sequence of cellular changes that would enable these cells to form a whole plant. (iv) <i>Agrobacterium tumefaciens</i> causes crown gall disease through rapid and uncontrolled cell division. Analyze how studying this disease mechanism led to a beneficial application in plant science. Answer: (i) Totipotency is the ability of some mature plant cells to undifferentiate, divide and redifferentiate to develop into a new plant under specific conditions. (ii) Condition X lacks air. Growth reduced because air is necessary for the cultured plant cells to grow and increase their fresh weight. (iii) The phloem cells first dedifferentiate, meaning they regain the ability to divide and form an undifferentiated mass of unspecialised cells. These unspecialised cells then divide and redifferentiate under appropriate conditions to form roots, shoot, and eventually the complete plant. (iv) Scientists studied how <i>Agrobacterium</i> transfers its genetic material into plant cells causing the disease. This knowledge was then used to employ <i>Agrobacterium</i> as a tool to introduce useful genes into plants for the production of valuable phytochemicals, improved crops, and disease-resistant varieties. ► Why? i) Totipotency is the ability of some mature plant cells to undifferentiate, divide and redifferentiate to develop into a new plant under specific conditions. ii) Condition X lacks air. Growth reduced because air is necessary for the cultured plant cells to grow and increase their fresh weight. iii) The phloem cells first dedifferentiate, meaning they regain the ability to divide and form an undifferentiated mass of unspecialised cells. These unspecialised cells then divide and redifferentiate under appropriate conditions to form roots, shoot, and eventually the complete plant. iv) Scientists studied how <i>Agrobacterium</i> transfers its genetic material into plant cells causing the disease. This knowledge was then used to employ <i>Agrobacterium</i> as a tool to introduce useful genes into plants for the production of valuable phytochemicals, improved crops, and disease-resistant varieties. [Ch 3: Tissues in Action]	5

